From Brick to Click: The Competitive and Welfare Effects of Digital Retail

Adrián Moreno María* Néstor Duch-Brown†

Abstract

Using panel data on vacuum cleaner sales across Belgium, Germany, and Slovakia from 2015 to 2019, we estimate a random coefficients demand model to quantify the competitive and welfare role of the online-only channel in a market traditionally served by click-and-mortar retailers. We differentiate between click-and-mortar retailers, including technological superstores, small retailers, and mass merchandisers, and online-only retailers, with marketplaces playing a significant role. Removing the online-only channel reduces consumer surplus by 12–17% and producer surplus by 7–10%. The online-only channel disciplines prices through stronger competition, benefiting consumers, while also expanding the overall market by reaching consumers who would otherwise not purchase at all, benefiting producers as well. When the online-only option is unavailable, consumers mainly shift toward retailers that replicate key features of online shopping, wide assortment, price transparency, and immediate product availability, rather than toward smaller, service-oriented stores. Our results highlight the importance of policies that sustain competition in the online environment, as they ensure that the welfare gains from e-commerce are shared across the market.

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*Departament d'Economia Aplicada, Universitat Autònoma de Barcelona, Barcelona, Spain. Email: adrian.moreno.maria@uab.cat

†Institute for Prospective Technological Studies (IPTS), Joint Research Centre, European Commission, Inca Garcilaso 3, 41092 Seville, Spain. Email: Nestor.DUCH-BROWN@ec.europa.eu

1 Introduction

The rapid expansion of e-commerce has transformed the competitive structure of retail markets. In Europe alone, more than 150,000 firms now sell goods and services online, a transition that accelerated substantially during the Covid-19 pandemic. Yet the growth of online commerce has not simply shifted transactions from physical stores to websites. It has also brought into direct competition two distinct retail models: online-only retailers, which operate without a physical store network, and click-and-mortar retailers, which combine an established physical presence with an online storefront.

This distinction is important because competition between these two retail models goes beyond a simple online-versus-offline comparison. Consumers purchasing from click-and-mortar retailers can benefit from both digital access and services associated with physical stores, such as presale advice, product inspection, immediate collection, or post-sale assistance. Online-only retailers, by contrast, operate without a physical store network and can exploit features such as broad geographic reach, large assortments, convenient search and comparison tools, and centralized logistics. The expansion of e-commerce has therefore created a competitive environment in which firms with very different retail technologies increasingly serve the same consumers.

The effect of online-only retailers on incumbent click-and-mortar firms is theoretically ambiguous. On the one hand, their presence may intensify competition and divert demand away from incumbent retailers, reducing their market shares and putting downward pressure on prices. On the other hand, the characteristics of online commerce may allow online-only retailers to reach consumers who would otherwise not purchase from any of the existing retail formats. In this case, their presence does not merely reallocate demand across retailers but also expands the effective size of the market. Distinguishing between these two mechanisms—business stealing and market expansion—is important for understanding both the competitive and welfare effects of e-commerce.

This paper studies this distinction by examining competition between online-only generalists—a category that in our data includes sales through major online marketplaces—and click-and-mortar retailers in the European market for household electronic appliances. Rather than asking whether consumers prefer purchasing online or offline per se, we ask how the presence of an online-only retail channel affects firms that already combine physical stores with an online presence. This distinction allows us to study whether online-only retailers primarily substitute for existing retailers, whether they bring additional consumers into the market, and which types of incumbent retailers face the strongest competitive pressure from their presence.

We address two main questions. First, do online-only retailers primarily reallocate demand away from click-and-mortar retailers, or do they expand the market? Second, what are the consequences of their presence for consumer and producer welfare?

We address these questions using detailed product-level sales data for the vacuum cleaner market in Belgium, Germany, and Slovakia over five years. The data cover sales through the online-only channel and three distinct click-and-mortar formats—independent retailers (e.g., buying groups such as *Euronics* and *Electronic Partner*), technological superstores (e.g., *Mediamarkt* and *Elektro Haas*), and mass merchandisers (e.g., *Metro* and *Interspar*)—allowing us to observe competition and substitution across retail formats within the same product market. Vacuum cleaners are a widely owned durable good with a relatively predictable replacement cycle, making them a useful setting for studying channel substitution and market expansion.

These click-and-mortar formats differ substantially in assortment and service provision. Evidence from the European Commission's E-commerce Sector Inquiry (2017)[1] highlights the importance of non-price dimensions—such as product expertise, presale advice, physical inspection, and post-sale services—in consumers' channel choices. Selective distribution practices in the electronics and appliance sector further contribute to these differences, as technological superstores typically offer broader assortments than mass merchandisers and smaller retailers. These differences provide a natural source of heterogeneity in how click-and-mortar formats compete with online-only retailers (Section 3.1 discusses this institutional background in more detail).

We estimate a random-coefficients demand model, following the approach proposed by Berry et al. (1995), that allows consumers to differ in both price sensitivity and their valuation of retail formats, generating flexible substitution patterns across channels. The model allows us to quantify how consumers reallocate purchases across retailers, how much demand shifts to the outside option, and the resulting implications for prices and welfare. We use the estimated model to construct a counterfactual in which the online-only channel is removed from the market—raising its marginal costs to a prohibitive level and re-solving for the Bertrand-Nash equilibrium—to recover how prices, market shares, and welfare would change in its absence.

Our demand estimates reveal substantial heterogeneity in consumers' valuation of retail channels. In particular, consumers with a stronger preference for the online-only channel tend to be less price-sensitive, suggesting that the appeal of online retail extends beyond lower prices and is consistent with consumers valuing convenience and reduced shopping costs. The

[1]European Commission (2017), *Commission publishes final report on the E-commerce Sector Inquiry*, available at: https://ec.europa.eu/commission/presscorner/detail/en/ip_17_1261.

counterfactual analysis further shows that the online-only channel does not merely reallocate demand away from incumbent retailers. When consumers do not consider the online-only option, a substantial fraction of those who would otherwise purchase through this channel choose the outside option rather than switching to a click-and-mortar retailer, indicating that online-only retail expands the effective size of the market. Among click-and-mortar formats, technological superstores absorb the largest share of diverted demand, substantially more than small retailers or mass merchandisers. This substitution pattern is consistent with technological superstores sharing several characteristics with online-only retail, including broad assortments and greater price transparency. In a supplementary specification, we model this relationship directly by allowing the random coefficients on consumers' tastes for the online-only and technological-superstore channels to be correlated. Although this is not our preferred specification, since identifying this covariance comes at the cost of dropping the random coefficient on price, we estimate a positive and significant correlation between the two channel tastes. This result is consistent with, and reinforces, the substitution pattern observed in the counterfactual analysis.

The online-only channel also strengthens competition in the market. When consumers do not consider this option, equilibrium prices increase and consumer surplus falls by approximately 12–17% across specifications. Aggregate producer surplus also declines, by around 7–10%, as the additional demand captured by click-and-mortar retailers does not fully compensate for the consumers who move to the outside option. Overall, the online-only channel plays a dual role by increasing competitive pressure and expanding market demand.

Our analysis contributes to the literature on e-commerce and cross-channel competition in two respects. First, while much of the existing literature studies substitution between online and offline purchasing, we distinguish between online-only retailers and heterogeneous click-and-mortar formats that themselves operate online. Second, our data allow us to quantify not only substitution across retail formats but also the extent to which online-only retail expands market demand, and the resulting consequences for consumer and producer welfare.

Our findings also speak to the ongoing regulation of digital markets in Europe. While restrictions imposed by manufacturers on where or how retailers may sell their products—including restrictions on the use of online marketplaces—are primarily assessed under EU competition rules governing manufacturer–retailer agreements, the Digital Markets Act (DMA) addresses a different but related issue: the competitive conditions under which large digital platforms operate. For designated gatekeeper marketplaces, the DMA limits practices such as self-preferencing and discriminatory ranking (Art. 6(5)) and restricts the use of non-public business-user data in competition with those users (Art. 6(2)). Our results do not provide a direct evaluation of these provisions, but they highlight why the competitive conditions

under which large online intermediaries operate can matter beyond the platform itself. By showing that the online-only channel disciplines retail prices and expands market demand, our counterfactuals suggest that policies affecting the contestability and accessibility of this channel may have welfare consequences throughout the broader retail market, including for consumers and competing click-and-mortar retailers. In this sense, vertical rules governing retailers' access to marketplaces and the DMA rules governing competition within gatekeeper platforms operate at different levels of the distribution chain, but both can ultimately affect the strength of the competitive constraint exerted by online retail.

2 Literature

This paper is directly related to the literature on the effects of e-commerce and cross-channel competition. Goolsbee (2001) examines substitution between online and offline sales of personal computers and finds that the cross-price elasticity is larger than one: when local prices rise, buyers tend to purchase computers online. Similarly, Prince (2007) conducts a detailed investigation into the evolving market dynamics of e-commerce in the retail sector, revealing that online and traditional retail channels for personal computers have transitioned into strong substitutes over time.

Wallsten (2013) and Gentzkow (2007) highlight a substitution trend toward online media, replacing traditional media consumption. Pozzi (2013) finds that online distribution channels increase overall revenues, with limited cannibalization of traditional sales. Conversely, Forman et al. (2009) analyze the impact of physical store availability on online purchase decisions, finding a preference for in-person shopping over online options. Brynjolfsson et al. (2009) find that the presence of local stores reduces the sensitivity of consumers to online price discounts for mainstream products, satisfying the consumer's need by offering instant gratification.

Chiou (2009) investigates how store reputation, perceived value, and shopping experience significantly influence consumer choices across channels. The study reveals a preference for Wal-Mart even when similar products are available through other offline and online channels. This finding aligns with Smith and Brynjolfsson (2001), who highlight brand reputation as an indicator of shipping reliability. Further, Brynjolfsson and Smith (2000) show that Internet retailers face significant competition from brick-and-mortar retailers in the mainstream product market, but less rivalry exists when selling niche products.

However, the literature has not addressed the impact of online-only players on the market, mostly focusing on sales made by click-and-mortar businesses that operate through both offline and online channels. Moreover, discussions on the welfare effects of e-commerce are

scarce. This oversight is significant given the fact that the reduction of search costs and the use of price comparison tools increase consumers' surplus (Brynjolfsson et al., 2003; Ellison & Ellison, 2018). These features not only improve the shopping experience but also boost consumer benefits.

Marketplaces, included among online-only options in the market, make use of these mechanisms. Consumers also benefit from market expansion and product variety (Ershov, 2018; Gentzkow, 2007). On the producer's side, the work is also scarce and mostly restricted to comparisons between offline and online sales channels without considering the diversity of business models (Duch-Brown et al., 2017). Producers also benefit from market expansion, which increases gross profits. However, the lack of detailed analysis regarding the nuances of online sales leaves a gap. The absence of detailed analysis on recent e-commerce advancements and marketplace dynamics further opens the path for our investigation.

At a moment where digital commerce has transitioned from a mere alternative to often being the primary option for consumers, our research significantly contributes to academic work. We closely follow the structural approach pioneered by Berry et al. (1995) to investigate the impact of previously unexplored aspects of e-commerce on the market. We offer insights into the adoption of online storefronts by traditional marketers—known as click-and-mortar channels—and the rise of online-only players, entities dedicated exclusively to online commerce where marketplaces play a major role.

As a result, we believe that our study contributes to a narrative of change, innovation, and continual development within the rapidly evolving digital economy.

3 Differences between Sales Channels

In today's retail industry, businesses operate across different channels, each fulfilling distinct consumer needs. We consider a complete view of the main sales channels in the market. The panel market is composed of four groups: small retailers, technological superstores, mass merchandisers, and online-only retailers (Figure 1).

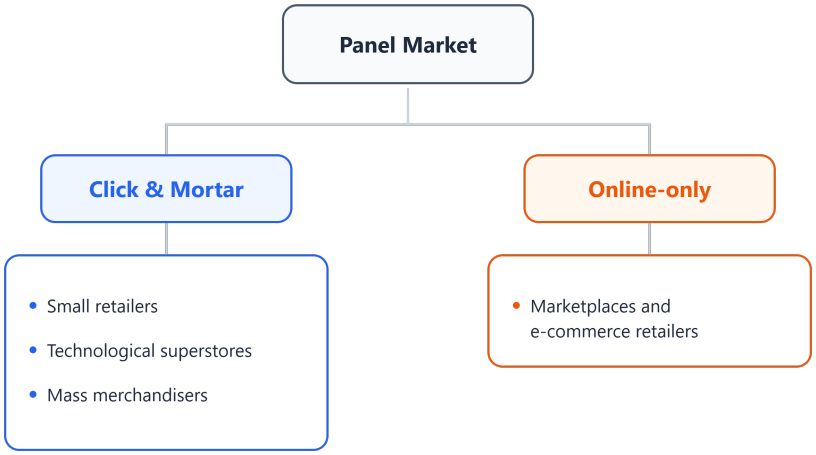


Figure 1: Panel Market: Main Sales Channels

Small retailers and buying groups. These are independently owned, local or regional consumer electronics stores that individually lack the scale of large chains. Many join retail alliances to gain purchasing power and brand support while keeping their own store identity: examples include buying groups such as *Euronics*, where independent retailers collaborate to obtain better supplier deals, and store networks such as *Electronic Partner*, which operate under a shared central brand with standardized practices. These arrangements let otherwise small, independently owned stores access some of the economies of scale that larger chains have on their own.

Technological superstores. Large-scale, specialized electronics retailers that operate through company-owned stores or franchised outlets, offering a wide range of consumer electronics and household appliances under a centralized brand. Examples include *Elektro Haas* or *Mediamarkt*.

Mass merchandisers. Large-scale retailers offering a wide assortment of products at competitive prices, targeting high sales volumes and broad consumer segments. This category includes *Cash & Carry* retailers like *Metro*, as well as hypermarkets and supermarkets such as *Interspar* or regional chains.

Online-only retailers. Retailers that operate exclusively online, with no physical storefronts. This category spans broad generalists carrying products across many categories—electronics, home appliances, fashion, books, and more—as well as online specialists focused on consumer electronics and household appliances. Their business model emphasizes scalability, cost efficiency, and logistics optimization to serve large consumer bases. Examples include *Amazon*, a broad generalist, and *0815* and *Redcoon*, online specialists in consumer electronics.

Among these online-only retailers, a significant portion adopt a marketplace model, where the platform intermediates between buyers and sellers rather than selling products directly. *Amazon*, already noted above, is an example. Marketplaces have become a dominant force by consolidating a vast number of sellers, making it easier for consumers to find and compare products. They offer advantages to sellers such as visibility and access to a larger consumer base. According to (Amazon, 2023), more than 60% of sales on its platform come from independent sellers. The decision to sell through marketplaces versus maintaining one's own e-commerce website is actively debated: marketplaces can drive significant traffic and sales, but may also induce cannibalization if the marketplace competes with its sellers. Prior work notes that marketplaces often become the starting point of consumers' shopping journeys, while other authors caution that sellers risk losing customer loyalty and control over the shopping experience.

Online-only retailers operate exclusively online, whereas other categories operate both in person and online (click-and-mortar). Some—particularly technological superstores—are steadily increasing their share of online sales, while smaller retailers continue to rely on traditional selling methods, gradually incorporating online channels. This shift toward digitalization reflects a broad trend in retail, where online-only shopping is becoming increasingly dominant. Operating exclusively online allows online-only retailers to reduce costs and offer competitive prices.

3.1 Institutional Background

Evidence from the European Commission's E-commerce Sector Inquiry (2017)[2] shows that the expansion of online commerce affects traditional distribution structures. The report documents different distribution strategies and consumer behaviour in durable goods, including consumer electronics and household appliances. It highlights greater price transparency, stronger interactions between online and offline channels, and the increasing relevance of service quality and brand positioning in shaping consumers' channel choices. Although e-

[2]European Commission (2017), *Commission publishes final report on the E-commerce Sector Inquiry*, available at: https://ec.europa.eu/commission/presscorner/detail/en/ip_17_1261.

commerce has made it easier for consumers to compare prices across retailers and channels, online and offline formats remain differentiated along key non-price dimensions. These include presale advice, product expertise, the ability to provide physical demonstrations, and post-sale services. Our empirical results are consistent with this view: consumers do not regard all channels as offering the same value. In particular, technological superstores provide a substantially wider range of products than other channels, whereas mass merchandisers carry the narrowest assortments. This difference in assortment is not incidental. The electronics and appliance sector makes widespread use of selective distribution systems, whereby manufacturers authorise only retailers that meet specific quality and service criteria to sell certain brands or product lines. Consistent with this, technological superstores, which typically satisfy these requirements, gain access to a broader range of authorised products, while mass merchandisers, which often do not meet these standards, show lower participation and carry more limited assortments. The report also shows that similar constraints apply within the online channel itself. More than 90% of retailers selling online rely on their own webshops, while only a small share uses marketplaces. In Belgium, for example, only 4% of retailers report selling through marketplaces, compared with 62% in Germany. Moreover, marketplace restrictions are widespread in sectors involving branded, technical, or complex goods. Around 18% of retailers report explicit contractual limitations on marketplace sales, and even higher figures in Germany (32%) and France (21%). Despite the relatively limited adoption of marketplaces by retailers in some countries, the sales volume generated through these platforms is substantial. Based on recent ECDB[3] data 72% of global e-commerce revenue stems from marketplaces, while in Europe the balance appears more even, with approximately 47% of online sales generated through marketplaces and 53% through own-webshop channels. However, in the cross-border European e-commerce, marketplaces account for around 70% of total turnover. Overall, despite uneven adoption and frequent contractual limitations, marketplaces play a large role in e-commerce sales volumes.

These institutional constraints have direct implications for channel substitution in our empirical setting. Because small and medium-sized retailers often rely on marketplaces to expand their reach online, marketplace bans prevent many independent retailers from offering their full product range across channels. Consistent with this, when either the online-only channel or the independent retail channel disappears in our counterfactual simulations, technological superstores absorb a significant share of diverted demand, reflecting their richer assortments and higher service levels.

[3]ECDB (2025), *Global eCommerce 2025*, available at: https://ecdb.com/whitepaper/global-ecommerce-2025.

4 Data

We use data collected by the *Gesellschaft für Konsumforschung* (GfK) on sales information for vacuum cleaners in several European countries. We analyze data spanning 2015–2019 from both offline and online markets.

GfK is a market research firm that compiles retail sales data through a panel of participating retailers spanning all major distribution formats—independent stores, retail chains, technological superstores, mass merchandisers, and online-only retailers. Participating retailers periodically report point-of-sale transaction records (units sold and revenue) for every product model they carry, which GfK aggregates and anonymizes into standardized, model-level panels by country, channel, and time period. Because the underlying source is retailer-reported point-of-sale transactions, each recorded sale corresponds directly to a consumer purchase, and by aggregating across participating retailers, the resulting figures approximate the full population of consumer purchases in each market.

The market is divided into sales from three distinct types of click-and-mortar retailers and online-only retailers, as explained in the previous section. This segmentation allows us to better understand the competition in the online market, particularly how different click-and-mortar retailers are affected by new online channels. The online-only category includes both independent online retailers and marketplaces; although our dataset does not include a detailed breakdown within this segment, marketplaces are known to account for a large and growing share of online sales.

The dataset contains information on 6,202 observations in Belgium, 4,724 in Germany, and 5,932 in Slovakia. We define a market as a combination of country and year, while a product is defined as a combination of item and channel. Market share for each product is calculated by dividing the total number of sales by the total potential number of products in a market.

For each observation, we have information on price, distribution channel, and features such as wattage, noise level (in dB), filter efficiency, power brush and power supply capacity, handheld control option, special animal option, and model. This allows us to map consumer preferences to specific components of a purchased vacuum cleaner.

Prices reflect only the final product costs, excluding any delivery charges. Therefore, prices for online products are those provided directly by the stores. If delivery charges apply, we are not able to account for them. As we do not have information about delivery costs, we are also unable to specify delivery type. This lack of information means our analysis focuses only on the base price of the product, without delivery fees. We acknowledge that delivery costs may increase final prices and are relevant for a comprehensive analysis; however, the

base price remains a critical factor in consumer decision-making and market competitiveness.

Additionally, each observation includes aggregate sales information—total units sold—summarized by country, year, and channel in Table 1. To improve precision in modeling consumer choices, we remove products with a market share below 0.001%, which reduces noise from long-tail products without materially affecting the sample: these products account for only 3,071 observations and 0.25% of total market share. The final dataset contains 16,858 observations.

To ensure realism, the model accounts for consumer preference for an outside good. We define the potential market size based on potential buyers. The literature usually estimates this at 20–80% of the population in the market. However, since a household usually purchases only one vacuum cleaner, we base our market size on the number of households. Ownership rates are generally high across the European markets we study: for example, 88% of consumers in Germany and 85% in Belgium report owning a vacuum cleaner.[4] Industry evidence points to a typical replacement cycle of approximately five years, rising to around eight years for certain upright and cylinder models.[5] Considering that vacuum cleaners are generally replaced every five years, we set the potential market to be 20% of total households.[6] Table 1 presents the distribution of observations by sales channel, offering a clear view of the market.

[4]Statista (2025), *Share of vacuum cleaner owners in selected countries worldwide.* Available at: https://www.statista.com/forecasts/1452561/share-of-vacuum-cleaner-owners-in-selected-countries-worldwide.

[5]Mintel (Mintel, 2013) reports that 44% of UK households purchased a vacuum cleaner over a three-year period—the second most commonly bought domestic appliance—with consumers expecting a lifespan of about five years. Hoover's own guide (n.d., *How long should a Hoover last?*, available at: https://www.hoover-home.com/en_GB/blogs/news/how-long-should-a-hoover-last) puts upright and cylinder models at around eight years and cordless models at about five. Consistent with this, 28% of vacuum cleaner purchases in 2012 replaced appliances under that age that had broken down or become unreliable (WRAP, 2013).

[6]We also verified that using alternative replacement rates consistent with industry evidence—6 to 8 years—does not affect our results.

Table 1: Average Sales by Country, Year, and Channel

Country	Year	Small Retailers	Tech Superstores	Mass Merchandiser	Online-only
BE	2015	323,685.5	240,877.0	70,934.0	42,762.5
	2016	315,435.0	232,519.0	89,383.0	62,850.2
	2017	318,099.6	232,539.8	76,792.7	68,543.5
	2018	299,057.2	222,273.5	57,327.0	83,751.2
	2019	250,005.6	194,917.7	46,689.0	103,151.5
	Avg.	301,092.5	225,083.1	67,425.7	73,398.6
GE	2015	278,601.0	1,216,466.0	891,713.1	743,996.4
	2016	218,151.5	1,134,459.0	961,199.9	970,286.6
	2017	238,344.7	1,192,197.0	903,456.4	984,779.7
	2018	194,038.3	1,081,482.0	862,634.4	1,050,912.0
	2019	173,303.9	1,073,682.0	715,042.6	984,014.1
	Avg.	219,307.7	1,140,312.0	868,535.9	956,674.3
SL	2015	29,704.9	95,898.8	9,166.0	31,228.3
	2016	30,207.4	97,972.3	12,270.2	39,949.4
	2017	29,745.2	102,858.0	14,074.1	44,827.0
	2018	27,195.1	98,079.2	9,184.3	51,592.0
	2019	23,205.9	95,357.6	9,514.4	47,869.8
	Avg.	28,132.7	98,026.7	11,174.3	43,566.5

5 Methodology

We model demand using a discrete-choice framework in which consumers choose among products sold through four channels—small retailers, technological superstores, mass merchandisers, and online-only retailers—or an outside option representing consumers who purchase other goods or do not purchase at all.

Our analysis benefits from yearly panel data from three countries, which enables us to estimate a random coefficients logit model following the approach proposed by Berry et al. (1995). This methodology allows for flexible substitution patterns across products by introducing consumer-level heterogeneity in preferences. Specifically, our random coefficients specifications incorporate a random coefficient for prices together with a random coefficient for the online-only channel, and, in one specification, extend this to a covariance between the online-only and technological-superstores channels, capturing how consumer sensitivity

to prices and channel preferences vary across individuals.

5.1 Demand Side

We present the demand model as a sequence of four specifications of increasing flexibility. Each specification relaxes a substitution restriction imposed by the previous one, and we motivate every step by the specific empirical question it allows us to answer that its predecessor could not. All four specifications share the same market and product definitions: denote market $t = 1, \ldots, T$ as the combination of a country and a year ($T = 15$ in our panel of three countries over five years), and let $j \in J_t$ denote a product available for purchase in market t, with 0 denoting the outside option, normalized as

$$U_{i0t} = \varepsilon_{i0t}. \tag{1}$$

Every specification also includes brand fixed effects $\xi_{b(j)}$, a market-product demand shock ξ_{jt} that subsumes country and year fixed effects,[7] and a vector x_{jt} of non-channel product characteristics (wattage and EPA filter efficiency), alongside vacuum cleaner type absorbed as a fixed effect.[8] What varies across the four models is the treatment of the distribution channel and the stochastic structure of consumer taste heterogeneity. We denote by $Online_{jt}$, $SupersChain_{jt}$, and $MassMerch_{jt}$ the dummy variables indicating that product j is sold, respectively, through the online-only channel, technological superstores, or mass merchandisers, with small retailers as the omitted baseline category.

Model 1: Nested Logit. The first relaxation of IIA groups products into nests defined by their distribution channel, allowing the idiosyncratic taste shock to be correlated within a nest[9]:

$$U_{ijt} = \delta_{jt} + \zeta_{ig(j)t} + (1 - \rho)\, \varepsilon_{ijt}, \tag{2}$$

where $g(j)$ denotes the channel-nest to which product j belongs, ζ_{igt} is a common taste shock shared by all products within nest g in market t, and $\rho \in [0, 1)$ is the nesting parameter

[7] We also tested including a linear year trend in place of, and alongside, the year fixed effects; the trend coefficient was not statistically significant, and its inclusion does not alter our results.

[8] We do not separately include special animal, handheld control, power supply, or noise level: these characteristics are largely determined by vacuum cleaner type (animal-specific attachments, power supply, and typical noise levels vary systematically and almost mechanically across handheld, upright, canister, and robot models), so once type is absorbed as a fixed effect they carry little independent variation and add no meaningful explanatory power.

[9] We also tested a two-level nested structure, with vacuum cleaner type as an additional nesting level alongside channel; the results did not support establishing the model this way, so we retain the single-level channel nest as our specification.

governing the correlation of tastes within a nest. The model yields the linear estimating equation

$$\ln(s_{jt}) - \ln(s_{0t}) = \alpha p_{jt} + \beta x_{jt} + \rho \ln\Big(s_{j|g(j)t}\Big) + \xi_{b(j)} + \xi_{jt}, \tag{3}$$

where $s_{j|g(j)t}$ is product j's share within its own nest.[10] If ρ is estimated to be well within the unit interval and strongly significant, this confirms that substitution within a channel is systematically stronger than substitution across channels: when a product exits the market, other products in the same nest absorb a disproportionate share of the diverted demand relative to their raw market share. The corresponding own-price elasticity,

$$\eta_{jj} = -\alpha p_{jt} \left[\frac{1}{1-\rho} - \left(\frac{\rho}{1-\rho} \right) s_{j|g(j)t} - s_{jt} \right], \tag{4}$$

reinforces substitution within a nest relative to the standard logit elasticity, since $1/(1-\rho) > 1$ scales up every product's own-price response whenever $\rho > 0$. We report and discuss the estimated $\hat{\rho}$ and the implied elasticities in Section 6. Nested logit still restricts substitution to be symmetric within and between nests: every product in the online-only nest substitutes with every product in, say, the technological-superstore nest at the same cross-elasticity, governed by the single parameter ρ. It cannot capture that price sensitivity or the preference for the online-only channel might vary across consumers in ways correlated with unobserved tastes, precisely the kind of heterogeneity that determines who leaves the market entirely versus who switches channels once the online-only option disappears.

Model 2: Independent Price and Channel Random Coefficients. We move to a random coefficients (BLP-type) logit, which relaxes IIA fully by allowing marginal utilities to vary across consumers[11]:

$$U_{ijt} = \alpha_i p_{jt} + \beta x_{jt} + \gamma^{SC} SupersChain_{jt} + \gamma^{TM} MassMerch_{jt} + \gamma_i^{On} Online_{jt} + \xi_{b(j)} + \xi_{jt} + \varepsilon_{ijt}, \tag{5}$$

with two random coefficients,

[10] In this specification, nests coincide with channels: each product's nest is simply its sales channel. Because of this, the channel dummies $Online_{jt}$, $SupersChain_{jt}$, and $MassMerch_{jt}$ are not included as separate covariates in Equation (3): the term $\rho \ln(s_{j|g(j)t})$ already varies by nest, so it already absorbs any channel-level effect. Adding channel dummies on top would ask the data to separately identify two things that carry exactly the same information here, which is not possible. The correlation in tastes within a channel is therefore captured entirely by ρ, not by a combination of channel dummies and nest shares.

[11] We also tested a random coefficients nested logit, combining Model 1's channel-nest structure with random coefficients; this specification did not work either, so we move directly to the fully unrestricted random coefficients logit.

$$\alpha_i = \alpha + \sigma_{price}\, v_i^p, \qquad \gamma_i^{On} = \gamma^{On} + \sigma_{On}\, v_i^{On}, \qquad v_i^p, v_i^{On} \overset{iid}{\sim} \mathcal{N}(0,1), \tag{6}$$

assumed independent of each other—that is, a diagonal Σ with only two nonzero entries, σ_{price} and σ_{On}. Market shares are obtained by integrating over the joint distribution of the random coefficients,

$$s_{jt} = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k \in J_t} \exp(\delta_{kt} + \mu_{ikt})}\, dF(v_i), \tag{7}$$

where δ_{jt} collects the mean-utility terms and, under Model 2, $\mu_{ijt} = \sigma_{price}\, p_{jt}\, v_i^p + \sigma_{On}\, Online_{jt}\, v_i^{On}$. The same expression for s_{jt} applies to every random-coefficients specification that follows, with μ_{ijt} redefined accordingly in each. This is our baseline BLP specification. Nested logit summarizes all within-channel taste correlation in a single parameter, ρ, shared by every consumer. It therefore cannot tell whether that correlation arises because some consumers are simply insensitive to price, or because some consumers have an especially strong idiosyncratic preference for the online-only channel specifically. Two distinct explanations that Model 2 can separate by giving each source of heterogeneity its own random coefficient. Model 2 allows both types of heterogeneity to coexist and to shape substitution patterns independently. Consumers with a high v_i^{On} are disproportionately likely to leave the market entirely (rather than switch channels) when the online-only option is removed, while consumers with a high v_i^p drive the dispersion in own-price elasticities that a single α cannot generate. By construction, v_i^p and v_i^{On} are drawn independently, which rules out any systematic relationship between a consumer's price sensitivity and their taste for the online-only channel. If, for instance, price-insensitive consumers are also the ones most attached to the convenience of online shopping, Model 2 cannot represent this and will misstate how the price increase generated by removing the online-only channel translates into welfare losses. Model 3 (below) extends Model 2 by allowing exactly this interaction, letting the price and online-only channel random coefficients correlate.

Model 3: Correlated Price and Channel Random Coefficients. We extend Model 2 by allowing α_i and γ_i^{On} to be correlated, using a Cholesky parameterization of the 2×2 covariance matrix:

$$\begin{pmatrix} \alpha_i \\ \gamma_i^{On} \end{pmatrix} = \begin{pmatrix} \alpha \\ \gamma^{On} \end{pmatrix} + L \begin{pmatrix} v_i^p \\ v_i^{On} \end{pmatrix}, \qquad v_i^p, v_i^{On} \overset{iid}{\sim} \mathcal{N}(0,1), \tag{8}$$

where L is the lower-triangular Cholesky factor of $\Sigma = LL'$, and the off-diagonal element $\sigma_{p,On}$ captures the covariance between the two random coefficients. Model 3 directly

tests whether consumers who are less sensitive to price systematically place a higher (or lower) value on the online-only channel—a joint heterogeneity pattern directly relevant to the counterfactual, since it determines whether the welfare loss from removing that channel, for the consumers most attached to it, comes mainly from losing their preferred shopping format or is instead compounded by high sensitivity to the resulting price increase. Model 3 is a particularly rich specification, since it jointly retains the price random coefficient and its covariance with the online-only-channel taste while remaining reliably identified.

Model 4: Correlated Channel Taste Heterogeneity. As a supplementary specification exploring substitution between physical channels, we replace the price random coefficient with a *covariance* between the online-only-channel taste and the taste for *SupersChain*, loading γ_i^{SC} onto the same underlying draw used for γ_i^{On}:[12]

$$\gamma_i^{SC} = \gamma^{SC} + \sigma_{On,SC}\, v_i^{On}, \tag{9}$$

so that Σ is a 2×2 matrix over (On, SC), with taste heterogeneity for *SupersChain* entirely spanned by the same factor that generates heterogeneity in the taste for *Online* rather than an independent source of variation; this specification does not include a random coefficient on price. This restriction is grounded in economic reasoning: *Online* and *SupersChain* are, a priori, similar channels in the sense relevant for this paper: both offer the wide assortment, price transparency, and immediate product availability associated with large-scale retail formats, in contrast to the smaller, service-oriented small-retailer channel. It is therefore plausible that the same latent consumer type—one with a taste for scale, variety, and self-service shopping—drives heterogeneity in the valuation of both channels jointly, even if it does not generate meaningful *independent* variation in the valuation of *SupersChain* alone. The data support exactly this restriction: the covariance term $\sigma_{On,SC}$ is identified and economically informative, while the corresponding independent variance is not. We should be honest that this comes at a real cost: dropping the price random coefficient to make room for this covariance is a genuine limitation, not a costless choice, since

[12]The natural next step from Model 3 would have been a fuller specification with independent random coefficients on price and on every channel, and then on their pairwise interactions as well. This turns out to be infeasible: incorporating that many random coefficients is not identified given our data. Even the more modest step of incorporating independent random coefficients for price, the *Online*, and *SupersChain* channels—three separate variances, without any covariance terms between them—already fails: the standard error on σ_{SC} is an order of magnitude larger than the point estimate, and the condition number of the resulting 3×3 covariance matrix approaches machine precision, indicating a numerically singular Σ. This is unsurprising given the identification requirements of BLP-type models: separately pinning down an extra free parameter in Σ from aggregate share data alone—without demographic instruments or simulated agent-level data—requires more cross-market variation than our panel of 15 markets (three countries $\times$ five years) can support.

it leaves Model 4 unable to characterize how price sensitivity itself varies across consumers. For the same identification reasons, further attempts beyond Model 4 produced the same singular-Σ, explosive-standard-error pattern, so we do not extend this covariance structure to *MassMerch* or other channels. We therefore stop here, treat Model 4 as a supplementary specification rather than our preferred one, and report its supply-side and counterfactual implications alongside Models 2 and 3 in Section 6.3.

5.2 Supply Side

We consider an oligopolistic supply side, as is common in the literature.[13] The approach fits our market, which is dominated by a few major players that hold significant market shares and engage in product differentiation through various features and branding efforts. This setup is advantageous for two reasons. First, given observed market shares, prices, and characteristics—along with the demand-side estimation—it is feasible to invert the equilibrium first-order conditions to estimate products' marginal costs. Second, the model is designed to solve for new equilibrium price vectors under counterfactual scenarios.

Firms are assumed to compete in a Bertrand-Nash fashion. Each firm f sets prices to maximize profits:

$$\pi_f = \sum_{j \in J_f} (p_j - mc_j) q_j(p) \tag{10}$$

where mc_j is the marginal cost of product j, $s_j(p)$ is the market share of product j, and $q_j(p) = M s_j(p)$, with M representing the market size. Solving the first-order conditions, any product sold by firm f will satisfy:

$$s_j(p) + \sum_{r \in J_f} (p_r - mc_r) \frac{\partial s_r(p)}{\partial p_j} = 0 \tag{11}$$

Let $\Delta(p)$ denote the $J \times J$ matrix of own- and cross-price derivatives, and Z^F the block-diagonal matrix indicating product ownership (where $Z^F_{jr} = 1$ if j and r belong to the same firm, 0 otherwise). We can rewrite the previous system in matrix form as:

$$s(p) + (Z^F \times \Delta(p))(p - mc) = 0 \tag{12}$$

Solving for marginal costs gives:

$$mc = p - \Delta(p)^{-1} s(p) \tag{13}$$

[13] See Petrin (2002) for further details.

From this equation, we compute the marginal costs involved in production using observed prices, market shares, and estimated elasticities from the demand system. The estimated costs are then used to perform counterfactual analyses. Specifically, we simulate a new equilibrium price vector following the removal of goods or sales channels from the market. This allows us to assess the impact of the online-only channel by comparing equilibrium outcomes with and without it.

6 Empirical Results

We estimate the four demand specifications introduced in Section 5 sequentially. We divide this section into three subsections for clarity: the first discusses identification, the second presents the demand estimation results, and the third analyses counterfactual scenarios to examine the impact of the online-only channel on both producers and consumers.

6.1 Identification

Price is endogenous. When a product receives a positive unobserved demand shock ξ_{jt}, both its market share and its equilibrium price rise, since the firm raises price in response to higher demand. This biases the OLS price coefficient. We correct for this using the instruments proposed by Berry et al. (1995): aggregate characteristics and product counts of all of a firm's other products and of its rivals' products, in the same market. The identifying logic is that, under Bertrand-Nash competition with differentiated products, a firm's optimal markup on product j depends on how close a substitute j is to its rivals—captured by rivals' *characteristics*, which shift equilibrium prices without themselves being components of product j's own unobserved appeal ξ_{jt}. This exclusion restriction, standard since Berry et al. (1995), requires that ξ_{jt} be uncorrelated with rival characteristics conditional on the brand, vacuum-cleaner-type, and market fixed effects we absorb; it would fail if, for instance, an unobserved market-wide shock were systematically correlated with the mix of characteristics carried by competing products in that market, but our market (country $\times$ year) fixed effects absorb exactly this kind of aggregate confound, leaving only idiosyncratic, product-specific variation in ξ_{jt} to identify α.

Given the relatively similar characteristics of some products in our market, we adopt an aggregation strategy similar to that used in nested logit applications. Verboven (1996) recommends aggregating the characteristics of products within each nest (and sub-nest) when constructing these instruments, to avoid diluting their power with rival products that are, by construction, poor substitutes. Our nested logit specification (Model 1) incorporates

nests directly, defined by distribution channel; the random coefficients specifications (Models 2–4) have no explicit nest structure, so we instead apply the same underlying rationale by aggregating the characteristics of all of a firm's products and those of its competitors, but only for different types of vacuum cleaners and retail channel—that is, we consider a rival's characteristics only when it belongs to a different vacuum-cleaner type and channel than the focal product. This approach mitigates the loss of instrument power that would result from treating near-identical products as informative rivals. Following Reynaert and Verboven (2014), we do not use these instruments directly in our reported estimates: we first fit the model with these instruments fixed, then construct approximate optimal instruments (Chamberlain, 1987) from that first fit, and report our final estimates using these optimal instruments. For our preferred specification (Model 3), the first-stage regression of price on these optimal instruments—controlling for the same market, brand, and vacuum-cleaner-type fixed effects as the demand equation—yields a joint F-statistic of 12.06 ($p < 0.001$), above the conventional threshold of 10 for strong instruments. The Hansen J statistic is effectively zero (one overidentifying restriction), which does not reject the model but carries limited power given how few overidentifying restrictions our four-instrument set provides; we report it for completeness rather than as strong evidence of correct specification on its own.

6.2 Demand Estimation

The results from the demand estimation are presented in Table 4, which reports coefficient estimates for Models 1–4 of the nested and random coefficients specifications introduced in Section 5. All four specifications are estimated on the same sample of 16,858 observations across the same 15 markets and share the same characteristic vector. We discuss each specification in turn below.

Model 1: Nested Logit. Column (1) of Table 4 reports the nested logit estimates, with channel as the nest (Equation (3)). Our specification yields a price coefficient of -1.078 and a nesting parameter of $\hat{\rho} = 0.574$, both statistically significant. The nesting parameter indicates that consumers' tastes for products sold through the same channel are meaningfully correlated. Substitution within a channel is noticeably stronger than substitution across channels. The implied average own-price elasticity is -3.55, consistent with the intra-nest reinforcement predicted by Equation (4).

Model 2: Independent Price and Channel Random Coefficients. Column (2) of Table 4 reports our baseline BLP specification, estimated with efficient (2-step) GMM. The

random coefficient on price is significant, $\hat{\sigma}_{price} = 0.836$, indicating heterogeneity in price sensitivity—some consumers are more price-sensitive than others. The random coefficient on the online-only channel is larger, $\hat{\sigma}_{On} = 3.331$, confirming substantial consumer heterogeneity in how the online-only channel is valued. Some consumers have a strong idiosyncratic preference for, or against, shopping online, independently of price.

Among mean coefficients, higher wattage reduces consumer utility ($\hat{\beta}_{watt} = -0.00105$), consistent with a broader preference for energy-efficient appliances and with the direction of EU regulatory pressure over our sample period: the applicable EU ecodesign regulation progressively tightened the maximum power input for most vacuum cleaner types, capping it at 900 watts for units placed on the market from September 1, 2017 onward, though not retroactively, and with categories such as robot vacuums exempt, consistent with our data still containing wattage values above 900 both before and after that date. Expressed as willingness to pay using the price coefficient as numeraire, $-\hat{\beta}_{watt}/\hat{\alpha}$ implies that consumers value a 100-watt reduction at roughly €1.8—and, integrated over the range of wattage observed in our data (900 to 2,600 watts), consumers are willing to pay up to €30 more for the lowest-wattage models than for the highest-wattage ones, comparable to the market's average markup. Consumers with an EPA filter derive higher utility ($\hat{\beta}_{epa} = 0.862$). Both effects are statistically significant. The mean channel coefficients reveal a clear ranking: technological superstores are valued significantly above small retailers ($\hat{\gamma}^{SC} = 0.619$), while both the online-only channel ($\hat{\gamma}^{On} = -2.953$) and mass merchandisers ($\hat{\gamma}^{TM} = -0.702$) are valued significantly below small retailers on average, though, for the online-only channel, this average masks the substantial heterogeneity captured by σ_{On} above.[14]

The model absorbs market (country × year) and brand fixed effects, capturing unobserved market-level fluctuations and brand loyalty, respectively—particularly relevant in online channels, where brand reliability and trust strongly influence consumer behavior.[15]

The implied average own-price elasticity is −7.50, larger in magnitude than under Model 1, as expected once consumer heterogeneity in price sensitivity and channel taste is allowed to shape substitution patterns. This pattern mirrors Grigolon and Verboven (2014), who show both analytically and in Monte Carlo experiments that a nested logit systematically underestimates the price coefficient (in absolute value) relative to a random coefficients logit whenever the omitted heterogeneity is real, their nesting parameter absorbs part of that

[14]Since vacuum cleaner type is absorbed as a fixed effect rather than reported as a coefficient, we do not report type-specific preferences for this model.

[15]We use brand rather than product (model) fixed effects: since *maxwattage* and *epafilter_gr* are fixed characteristics of each individual model, absorbing product-level fixed effects would leave these characteristics perfectly collinear with the fixed effect, making $\hat{\beta}_{watt}$ and $\hat{\beta}_{epa}$ unidentifiable. Brand fixed effects are coarse enough to preserve within-brand variation in wattage and filter status, while still controlling for unobserved brand-level quality and loyalty.

heterogeneity and mechanically rescales every utility parameter, including price. We take this as independent external evidence that the gap reflects a well-documented structural feature of comparing these two model classes, not an anomaly specific to our data or estimation.[16]

Using these estimates, we recover marginal costs and markups from the supply-side first-order conditions implied by Bertrand-Nash pricing. The mean marginal cost is €121.5, and the mean markup is 18.4% of price; see Table 8 for the breakdown by channel and Figure 3 for their full distributions.

Model 3: Correlated Price and Channel Random Coefficients. Column (3) of Table 4 extends Model 2 by allowing the price and online-only-channel random coefficients to correlate. The random coefficient on price remains significant and somewhat larger than under Model 2, $\hat{\sigma}_{price} = 1.097$, and the covariance term is positive and significant, $\hat{\sigma}_{p,On} = 1.024$. Because α_i enters utility with a negative mean effect, a positive covariance between α_i and γ_i^{On} means that consumers whose price coefficient is less negative—i.e., less price-sensitive consumers—systematically place a higher value on the online-only channel. This directly answers the question Model 2 could not: rather than price sensitivity and channel taste being independent sources of heterogeneity, the consumers most attached to the online-only channel are, on average, also the least price-sensitive. This has a specific welfare implication. For this group, the loss from removing the online-only channel comes mainly from losing their preferred shopping format itself—their utility is largely insensitive to the price increase that follows, since price barely enters their decision to begin with. A plausible economic explanation is the opportunity cost of time: consumers who are less price-sensitive tend to place a higher value on their time, and the online-only channel economizes precisely on the time cost of shopping—traveling to a store, comparing options in person, waiting in line—which is consistent with vacuum cleaners being an infrequent, already-decided purchase (Section 4) for which convenience is likely to dominate price-hunting. This is not the only plausible sign *a priori*: one could instead expect the online-only channel to attract more price-sensitive bargain hunters, since it also facilitates price comparison, which would imply a *negative* covariance between α_i and γ_i^{On}. That we estimate the opposite sign suggests that, in this market, the time saved by shopping online outweighs the price-comparison motive. The implied heterogeneity in the online-only-channel taste itself, once this covariance is accounted

[16] In their European car market application, the nested logit's price coefficient (-1.00) is 5.5 times smaller in magnitude than the random coefficients logit's (-5.52)—almost exactly the 5.5× ratio we obtain between Model 1 ($\hat{\alpha} = -1.078$) and Models 2/3 ($\hat{\alpha} = -5.878/-6.042$) here, in an unrelated market. Part of this gap is a mechanical rescaling. Grigolon and Verboven (2014) show that dividing the nested logit's price coefficient by $(1-\hat{\rho})$ moves it substantially closer to the random coefficients estimate. The same diagnostic here, $\hat{\alpha}/(1-\hat{\rho}) = -1.078/(1-0.574) = -2.53$, closes roughly 30% of the gap to Model 2's $\hat{\alpha} = -5.878$ (from a 5.5× to a 2.3× ratio)—so part, but not all, of the discrepancy is this mechanical rescaling.

for, is $\hat{\sigma}_{On} = 1.054$, not statistically significant.

Among mean coefficients, the pattern mirrors Model 2 and all are statistically significant: higher wattage reduces utility ($\hat{\beta}_{watt} = -0.0010$), an EPA filter raises it ($\hat{\beta}_{epa} = 0.834$), technological superstores are valued above small retailers ($\hat{\gamma}^{SC} = 0.640$), and mass merchandisers below ($\hat{\gamma}^{TM} = -0.691$); the mean coefficient on the online-only channel, however, is no longer statistically distinguishable from zero ($\hat{\gamma}^{On} = -0.923$), consistent with more of its explanatory power now being absorbed by the covariance with price sensitivity rather than by the mean utility term. The implied average own-price elasticity is -7.42, essentially unchanged from Model 2's -7.50. As a further, purely internal check, we split our own products into price quartiles and compare how the average own-price elasticity grows from the cheapest to the most expensive quartile under Model 1 versus Model 3 (Table 11). Under Model 1, the elasticity grows almost exactly proportionally with price (within 0.1% of proportional growth in every quartile); under Model 3, growth is increasingly sub-proportional, falling to 96.7%, 90.9%, and 79.6% of proportional growth in the second, third, and fourth quartiles, respectively—the same qualitative pattern, and a similar order of magnitude, as Grigolon and Verboven (2014) report for the car market (own-price elasticities rising by a factor of about 4× from the cheapest to the most expensive segment under their nested logit, versus about 2.2–2.3× under their random coefficients logit). This confirms, with our own products rather than only with external literature, that consumers who are less sensitive to price disproportionately purchase higher-priced vacuum cleaners under Model 3—exactly the mechanism that produces a smaller-magnitude price coefficient under Model 1.

The same supply-side conditions imply marginal costs and markups close to Model 2's: mean marginal cost €120.7, mean markup 18.9% of price (Table 8, Figure 4), consistent with the two specifications agreeing on the supply side while differing mainly in how price sensitivity and channel taste covary.

Model 4: Correlated Channel Taste Heterogeneity. Column (4) of Table 4 reports this supplementary specification. Jointly estimating a price random coefficient alongside an Online-only–Superstores covariance term is not numerically stable in our data—with only 15 markets and no individual-level demographic instruments to discipline the mixing distribution, the resulting covariance matrix approaches singularity (Section 5)—so this specification drops the random coefficient on price, a meaningful limitation given how central price heterogeneity has proven to be in Models 2 and 3, in order to still test whether consumer tastes for the online-only channel and technological superstores are correlated. Both random coefficients are large and precisely estimated: the standard deviation of the online-only-channel taste is $\hat{\sigma}_{On} = 2.622$ and that of the technological-superstore taste is $\hat{\sigma}_{SC} = 2.136$, and the

covariance between the two is positive and significant, $\hat{\sigma}_{On,SC} = 3.897$. As discussed in Section 5, we read this as evidence of a shared latent consumer type—one with a taste for the scale, variety, and self-service format common to both large-format channels—rather than as two independent sources of heterogeneity.

Among mean coefficients, the (linear, non-random) price coefficient is $\hat{\alpha} = -2.361$, statistically significant; watts and EPA filter enter with the same sign as in Models 2 and 3 ($\hat{\beta}_{watt} = -0.0006$; $\hat{\beta}_{epa} = 0.326$), both significant. The mean channel coefficients are all smaller in magnitude than in Models 2 and 3 once the corresponding random coefficients absorb most of the channel-taste variation ($\hat{\gamma}^{On} = -1.526$; $\hat{\gamma}^{SC} = -0.157$, not statistically significant; $\hat{\gamma}^{TM} = -0.243$). The implied average own-price elasticity is -3.32, about half the magnitude of Models 2 and 3's $-7.50/-7.42$—a direct consequence of the smaller linear price coefficient once price sensitivity is no longer separately randomized, and a reminder that this specification's aggregate welfare and elasticity magnitudes are not directly comparable to Models 2 and 3's (Section 6.3).

Marginal costs come out substantially lower here, and markups correspondingly higher, than under Models 2 and 3: mean marginal cost €96.2, mean markup 44.3% of price (Table 8, Figure 5)—a mechanical consequence of the smaller implied elasticity.

6.3 Impact of the Online-only Distribution Channel

6.3.1 Methodology and Base Equilibrium Levels

We perform a counterfactual analysis based on the estimated demand model. Specifically, we simulate the exit of online-only players and marketplaces from the market: by raising the online-only channel's marginal costs to a prohibitive level and re-solving for the Bertrand-Nash equilibrium, its products are effectively priced out of the market, allowing us to recover the equilibrium prices and market shares that would prevail under the absence of online-only players. This exercise allows us to quantify the effect of the online-only channel on market competition and welfare. We report this exercise under all three of our estimated random-coefficient specifications—Model 2 (RC v1), Model 3 (RC v2), and Model 4 (RC v3)—to assess how sensitive the welfare conclusions are to the random-coefficient structure assumed.

Before turning to the changes induced by the online-only channel's exit, it is worth flagging that the model-implied *base* (pre-exit) equilibrium levels of consumer surplus and producer profits are not directly comparable across specifications. Total consumer surplus at the base equilibrium is €684.0 million under Model 2, €788.4 million under Model 3, and €1,442.0 million under Model 4; total producer profits are €391.2 million, €402.1 million, and €905.1 million, respectively. Models 2 and 3 are of broadly comparable magnitude, but

Model 4's levels are 80–130% higher. This follows mechanically from the price coefficient: $\hat{\alpha} = -2.361$ under Model 4, versus -5.878 and -6.042 under Models 2 and 3 (Table 4), roughly half as large in magnitude—likely itself a consequence of Model 4 not including a random coefficient on price: without that channel to absorb cross-consumer heterogeneity in price sensitivity, the single mean coefficient must average over the whole population's response, which can pull its estimated magnitude down relative to Models 2 and 3, where that heterogeneity is instead split between the mean and the variance. In logit-type demand systems consumer surplus scales approximately as $1/|\alpha|$, so a smaller price coefficient mechanically inflates the implied welfare levels without reflecting any genuine difference in market outcomes. Absolute euro levels should therefore not be compared across specifications with substantially different $\hat{\alpha}$; the percentage changes reported below, which normalize by each model's own base level, remain the relevant metric for comparing the welfare impact of the online-only channel's exit across specifications.

6.3.2 Diversion Ratios and Welfare Effects

Model 3's implied marginal costs and markups are close to Model 2's (mean marginal cost €120.7 versus €121.5; mean markup 18.9% versus 18.4% of price), consistent with the two specifications agreeing on the supply side while differing mainly in how price sensitivity and channel taste covary. Model 4's are substantially different (mean marginal cost €96.2; mean markup 44.3% of price), a direct consequence of its much smaller implied own-price elasticity (-3.32 versus $-7.50/-7.42$ under Models 2 and 3); see Table 8 for the full breakdown by channel and Figures 4–5 for the full distributions.

We compute diversion ratios from the removed channel to the remaining ones. Diversion ratios capture substitution patterns between products and indicate how much of the lost demand from one product (or channel) is absorbed by others. Table 9 and Figure 2 report these equilibrium diversion ratios for all three specifications. Under Model 2, of the demand lost when the online-only channel exits, 23.3% is diverted to technological superstores, 12.9% to independent retailers/buying groups, and 8.2% to mass merchandisers, while the remaining 55.6% is lost to the outside option entirely. Under Model 3, the pattern shifts more toward physical substitution and less toward exiting the market entirely: technological superstores absorb 33.0%, independent retailers/buying groups 16.9%, and mass merchandisers 10.2%, with only 40.0% lost to the outside option. Under Model 4, the pattern is more concentrated still: technological superstores absorb 38.2%—more than under either Model 2 or Model 3—while independent retailers/buying groups (9.1%) and mass merchandisers (6.0%) absorb less, and 46.7% is lost to the outside option, between Model 2's and Model 3's. Across all three specifications, a substantial share of demand is consistently lost to the outside option

rather than switching to a physical channel—i.e., these consumers stop buying a vacuum cleaner altogether—consistent with the online-only channel expanding the overall market rather than merely cannibalizing physical retailers.[17]

Table 7 reports the changes in consumer welfare, calculated following the formulation in Appendix C. Removing the online-only channel causes aggregate consumer surplus to fall by 17.4% (€−181.1 million) under Model 2, by 12.1% (€−148.7 million) under Model 3, and by 14.7% (€−324.3 million) under Model 4. In all three specifications, the loss is by far the largest in Germany (€−162.8 million under Model 2, €−134.5 million under Model 3, €−291.6 million under Model 4), reflecting its much larger household population, followed by Slovakia and Belgium. Prices increase as competitive pressure decreases under every specification, indicating that the online-only channel exerts a strong disciplining effect on market prices regardless of the random-coefficient structure assumed: the removal of this channel consistently reduces competition, increases firms' market power, and raises prices. The online-only channel thus contributes positively to consumer welfare across all three specifications, primarily through lower prices and higher accessibility, alongside the convenience, variety, and comparison capabilities of online marketplaces.

On the producer side, Table 6 shows that removing the online-only channel decreases total producer surplus (profits) by 10.3% (€−61.6 million) under Model 2, 7.2% (€−42.3 million) under Model 3, and 9.1% (€−127.5 million) under Model 4, with the same cross-country pattern as consumer surplus in every case (losses concentrated in Germany, then Slovakia, then Belgium). That both consumer *and* producer surplus fall together under all three specifications—rather than one gaining at the other's expense—indicates that the online-only channel expands the overall market by attracting consumers who would otherwise remain outside it: without this channel, producers lose access to those consumers, leading to a contraction in market size and profits, not merely a reallocation of surplus. At the channel level, all three surviving physical channels gain profit, as they absorb part of the online-only channel's diverted demand even though the market contracts overall; technological superstores capture the largest share of that gain under Model 4.

This reallocation toward technological superstores is not a coincidence of the point estimates. It follows directly from the positive, statistically significant covariance $\hat{\sigma}_{On,SC} = 3.897$ introduced in Model 4 (Table 4), which lets the taste for the online-only channel and the taste for technological superstores be driven by the same underlying consumer type. Under Model 2 and Model 3, the random coefficient on the online-only channel is independent of

[17] This figure aggregates over all 15 markets. A country- and brand-level breakdown was computed under an earlier, pre-N=16,858 specification; recomputing it under the current sample and across all three random-coefficient specifications (Models 2–4) remains future work.

(Model 2) or correlated only with price (Model 3), so a consumer's valuation of technological superstores plays no special role in determining where the online-only channel's demand goes once it exits; diversion in these two specifications is shaped mainly by relative mean utilities and market shares, in the same spirit as nested-logit substitution. Model 4's positive covariance instead reveals that consumers with an especially strong idiosyncratic preference for shopping online also tend to value technological superstores highly. This is plausible because both channels offer the wide assortment, price transparency, and self-service format that a smaller, service-oriented retailer does not (Section 5). Therefore, when the online-only channel disappears, this same subset of consumers is disproportionately likely to relocate to technological superstores rather than to independent retailers, mass merchandisers, or the outside option. A diversion ratio built only from mean utilities, as in Models 2 and 3, cannot generate this pattern by construction.

We flag this result as suggestive rather than conclusive. Model 4 has no random coefficient on price (Section 5), so it does not jointly characterize how price sensitivity and channel tastes co-vary, and we cannot rule out that a specification combining both a price random coefficient and the Online-only–Superstores covariance would attenuate or sharpen this reallocation; this caveat should make us cautious about over-interpreting the exact magnitude of the shift. We attempted richer specifications along these lines and found them numerically unidentified: with only 15 markets and no agent-level demographic instruments to discipline the mixing distribution, adding further free parameters to the covariance matrix (Σ) pushes it toward singularity and standard errors toward infinity, as already discussed in Section 5. Model 4's single covariance term is thus the richest structure our data and instrument set can support in this case. It gives information about the direction of channel–taste correlation, but not the final word on it.

6.4 Robustness to the Assumed Replacement Cycle

Our baseline market size $M_t = \text{households}_{ct}/L$ assumes a uniform vehicle-cleaner replacement cycle of $L = 5$ years across all 15 markets. We check how sensitive Model 1 (Nested Logit), Model 2 (RC v1), and Model 3 (RC v2) are to this assumption under uniform changes in L.

Table 10 reports the price coefficient ($\hat{\alpha}$), the nesting parameter ($\hat{\rho}$), and the implied own-price elasticity for Model 1, together with the price coefficient, the standard deviation of the price random coefficient ($\hat{\sigma}_{price}$), and the implied elasticity for Models 2 and 3, across the baseline and three moderate departures from it: uniform $L = 3$, $L = 4$, and $L = 6$. Model 1's $\hat{\alpha}$, $\hat{\rho}$, and elasticity are numerically invariant to four decimal places in every scenario we tried—expected, since a uniform rescaling of M_t is absorbed by the market

fixed effects that identify these nested-logit parameters. Model 2's $\hat{\alpha}$ ranges more widely across scenarios (-4.871 to -7.491) than Model 3's (-5.727 to -6.178), a further sense in which our preferred specification is the more stable one; the Model 2 own-price elasticity nonetheless stays within 12% of the baseline (-6.65 to -8.25, versus -7.50). Model 2's $\hat{\sigma}_{price}$ is significant at conventional levels under the baseline and uniform $L = 3/L = 4$ ($t = 2.06$ and $t \approx 2.12$–2.13, respectively), but loses significance under uniform $L = 6$ ($t = 1.11$), even though the implied elasticity itself remains stable throughout. Model 3—our preferred specification—is more robust on this dimension: $\hat{\sigma}_{price}$ stays significant across every scenario we tried, including uniform $L = 6$ ($t = 2.23$–4.29), with the implied elasticity likewise stable and close to baseline throughout (uniform $L = 4$: 1.128, $t = 3.94$, elasticity -7.18; uniform $L = 6$: 0.913, $t = 4.26$, elasticity -7.21; uniform $L = 3$: 0.735, $t = 2.23$, elasticity -7.15).

By contrast, more aggressive attempts to close the Belgium–Germany/Slovakia s_0 gap entirely (Belgium $L \leq 3$, or a joint scenario with Germany/Slovakia at $L = 4$) push Model 2 into poorly identified territory: the price coefficient ranges from -1.60 to -21.88 across these scenarios, $\hat{\sigma}_{price}$ occasionally flips sign and loses significance altogether, and the implied elasticity swings between -1.45 and -16.69—an order of magnitude apart. We do not view these more aggressive scenarios as informative about the true model, and exclude them from Table 10 (Appendix A); we report them here only to delineate where the assumption stops being a reasonable normalization and starts driving the results.

7 Conclusions

E-commerce, characterized by personalized browsing experiences and tailored product assortments, offers consumers access to a vast variety of products. The emergence of click-and-mortar strategies, online-only retailers, and marketplaces has diversified the retail landscape, leading to the formation of new market segments and fundamentally transforming the way products are marketed and purchased.

This study investigates the competitive and welfare role of online-only players in the market. To do so, we employ a demand model of differentiated products that incorporates consumer preferences for distribution channels. Using a rich dataset on electronic device sales—including prices, product characteristics, and sales channels—we are able to disentangle the effects of different types of retailers. The dataset distinguishes between sales made by small retailers, technological superstores, mass merchandisers, and the online-only channel.

Our results indicate significant heterogeneity in consumer price sensitivity and distinct valuations of sales channels. In line with economic theory, we find that the presence of online-only retailers intensifies competition, leading to lower prices. The market expansion

driven by online-only players also enables firms to reach new consumers, which can increase overall profits.

To explore the implications of removing online-only players, we compute diversion ratios that reveal country-level differences. In Belgium, where small retailers dominate, consumers tend to rely on familiar shopping formats in the absence of the convenience and accessibility provided by online-only players. In contrast, in Germany and Slovakia, a significant share of consumers shifts to technological superstores, underscoring the importance of variety and strategic marketing—features that those stores share with online-only players. These findings highlight structural and behavioral differences across countries. On the consumer side, the presence of the online-only channel substantially enhances welfare by providing competitive prices, convenience, and greater accessibility.

Future research will aim to incorporate microdata on the supply side to better capture the dynamics between producers and consumers in retail markets. Integrating supply-side data would provide deeper insights into pricing strategies, firm behavior, and market equilibrium, thus enriching our understanding of the interaction between supply and demand in digital retailing.

Finally, based on our results, we emphasize the importance of preserving competitive, contestable access to the online-only channel. Our finding that removing this channel reduces both consumer and producer welfare simultaneously—rather than reallocating surplus from one to the other—suggests that its presence benefits the market as a whole, not consumers at sellers' expense. This is directly relevant to the EU's Digital Markets Act (DMA), discussed in Section 1: while our results do not evaluate the DMA's provisions directly, they show why the competitive conditions under which large online intermediaries operate can matter beyond the platform itself, since changes in the competitive position of online retail can affect competition and welfare throughout the broader retail market. Reducing entry barriers and enhancing market contestability in the marketplace sector can foster a more competitive environment, preventing market dominance by a few large players. Moreover, policies that encourage physical stores to adopt digital strategies—such as supporting click-and-mortar initiatives that combine online convenience with in-store experiences—can help traditional retailers remain competitive in an increasingly digital marketplace.

Appendix A. Additional Tables

Table 2: Summary Statistics (N = 16,858)

Variable	**Obs**	**Mean**	**Std. Dev.**	**Min**	**Max**
Prices	16,858	140.69	78.88	20.04	359.84
Wattage	16,858	826.62	444.43	1	2600
Watts dummy	16,858	0.95	0.62	0	2
Noise level	16,858	75.54	7.80	9	120
Power supply	16,858	0.13	0.33	0	1
Epa filter	16,858	0.52	0.50	0	1
Special animal	16,858	0.12	0.32	0	1
Type vacuum	16,858	0.41	0.79	0	3

Table 3: Distribution of products across channels by brand type (percentages).

	Small Retailers	Technological Superstores	Mass Merchandisers	Online-only
Non-top brands	16.3%	28.6%	20.8%	34.3%
Top brands	25.3%	36.3%	13.8%	24.6%

Table 4: Estimation Results: Models 2–5

	(1) Nested Logit	(2) RC v1	(3) RC v2	(4) RC v3
Panel A: Mean Coefficients (β)				
Price (α)	-1.078**	-5.878**	-6.042***	-2.361***
	(0.423)	(2.964)	(1.664)	(0.500)
Watts	-0.0002**	-0.0011***	-0.0010***	-0.0006***
	(0.0001)	(0.0004)	(0.0002)	(0.0001)
EPA filter: No	0.176**	0.862*	0.834***	0.326***
	(0.070)	(0.462)	(0.258)	(0.090)
Channel: Online-only (γ^{On})	—	-2.953**	-0.923	-1.526*
		(1.372)	(0.570)	(0.873)
Channel: Tech. Superstores (γ^{SC})	—	0.619***	0.640***	-0.157
		(0.085)	(0.075)	(0.288)
Channel: Mass Merchandisers (γ^{TM})	—	-0.702**	-0.691***	-0.243***
		(0.358)	(0.208)	(0.077)
Panel B: Nonlinear Parameters				
ρ (nesting parameter)	0.574***	—	—	—
	(0.124)			
sd Price (σ_{price})	—	0.836**	1.097***	—
		(0.405)	(0.256)	
sd Channel: Online-only (σ_{On})	—	3.331***	1.054[a]	2.622***
		(1.135)	(0.773)	(0.502)
sd Channel: Tech. Superstores (σ_{SC})	—	—	—	2.136***[b]
				(0.555)
Cov(Price, Online-only) ($\sigma_{p,On}$)	—	—	1.024***[a]	—
			(0.379)	
Cov(Online-only, Superstores) ($\sigma_{On,SC}$)	—	—	—	3.897**[b]
				(1.660)
Market FE	Yes	Yes	Yes	Yes
Brand FE	Yes	Yes	Yes	Yes
Vacuum cleaner type FE	Yes	Yes	Yes	Yes
Observations	16,858	16,858	16,858	16,858
Mean own-price elasticity	−3.55	−7.50	−7.42	−3.32
Weighting matrix	Robust	Robust	Robust	Robust

Standard errors in parentheses (robust).
*** p<0.01, ** p<0.05, * p<0.1. RC = Random Coefficients.
[a] Model 3 (RC v2) introduces a covariance between the price and Online-only random coefficients. The reported sd for Online-only and the covariance entry come from the implied covariance matrix $\Sigma = \text{Sigma} \times \text{Sigma}^{\top}$ (pyblp's "Sigma Squared"), not the raw Cholesky factor, since the latter's diagonal for the second random coefficient in the ordering omits the cross-term's contribution to its variance. The standard error for the sd entry is obtained via the delta method from the standard error of the corresponding variance term; the covariance entry and its standard error are read directly off Sigma Squared.
[b] Model 4 (RC v3, a supplementary specification) introduces a covariance between the Online-only and Superstores random coefficients instead of an independent variance for Superstores, which is not separately identified (see Section 5); this specification has no price random coefficient.

Table 5: Initial Market Outcomes

	Belgium	Germany	Slovakia
Average price (€)			
Small Retailers	162.2	160.4	93.8
Technological Superstores	156.9	142.3	113.1
Mass Merchandisers	114.5	111.1	72.2
Online-only	145.5	142.6	97.0
Consumer Surplus			
Per unit (€)	29.8	12.5	11.7
Total (€million)	145.7	512.4	25.9
Producer profits			
Total (€million)	65.3	308.7	17.2

Table 6: Change in Producer Profits

	(2) RC v1		(3) RC v2		(4) RC v3	
Country	Loss (M€)	Change (%)	Loss (M€)	Change (%)	Loss (M€)	Change (%)
Belgium	-2.03	-3.1%	-0.98	-1.5%	-4.43	-2.9%
Germany	-57.05	-18.5%	-39.21	-12.3%	-117.99	-16.6%
Slovakia	-2.57	-14.9%	-2.11	-12.0%	-5.11	-12.4%
Total	**-61.6**	**-10.3%**	**-42.3**	**-7.2%**	**-127.5**	**-9.1%**

Per-country % change is Loss/Initial for that country. All figures: $N = 16{,}858$.

Table 7: Change in Consumer Surplus

	(2) RC v1		(3) RC v2		(4) RC v3	
Country	Loss (M€)	Change (%)	Loss (M€)	Change (%)	Loss (M€)	Change (%)
Belgium	-11.43	-7.8%	-8.71	-5.0%	-20.42	-7.3%
Germany	-162.81	-31.8%	-134.53	-22.9%	-291.58	-26.6%
Slovakia	-6.90	-26.6%	-5.42	-19.7%	-12.25	-18.4%
Total	**-181.1**	**-17.4%**	**-148.7**	**-12.1%**	**-324.3**	**-14.7%**

Per-country % change is Loss/Initial for that country. All figures: $N = 16{,}858$.

Table 8: Marginal Costs and Markups by Channel (Base Equilibrium)

	(2) RC v1		(3) RC v2		(4) RC v3	
Channel	MC (€)	Markup (%)	MC (€)	Markup (%)	MC (€)	Markup (%)
Independent retailers / buying groups	126.14	17.84%	126.19	17.55%	101.48	42.01%
Online-only	119.01	18.65%	116.47	20.72%	93.24	45.74%
Technological superstores	125.82	17.88%	125.83	17.62%	100.16	43.06%
Mass merchandisers	110.35	19.95%	110.56	19.45%	85.54	47.92%

All specifications: $N = 16{,}858$.

Table 9: Equilibrium Diversion Ratios from the Simultaneous Exit of the Online-only Channel

	(2) RC v1	(3) RC v2	(4) RC v3
Technological Superstores	23.3%	33.0%	38.2%
Independent retailers / buying groups	12.9%	16.9%	9.1%
Mass Merchandisers	8.2%	10.2%	6.0%
Outside good	55.6%	40.0%	46.7%

Share of the demand lost when the online-only channel exits simultaneously (all its products removed at once) that is absorbed by each remaining channel in the new equilibrium, averaged across all 15 markets; see Section 6.3.

Table 10: Robustness to the Assumed Replacement Cycle L

	Model 2 (RC v1)			Model 3 (RC v2)		
Scenario	$\hat{\alpha}$ (SE)	$\hat{\sigma}_{price}$ (t)	Elasticity	$\hat{\alpha}$ (SE)	$\hat{\sigma}_{price}$ (t)	Elasticity
Baseline ($L = 5$, all countries)	−5.878 (2.964)	0.836 (2.06)	−7.50	−6.042 (1.664)	1.097 (4.29)	−7.18
Uniform $L = 6$	−4.871 (2.393)	0.420 (1.11)	−6.65	−5.788 (1.599)	0.913 (4.26)	−7.21
Uniform $L = 4$	−6.781 (3.634)	1.157 (2.13)	−8.07	−6.178 (1.728)	1.128 (3.94)	−7.18
Uniform $L = 3$	−7.491 (4.097)	1.449 (2.12)	−8.25	−5.727 (1.698)	0.735 (2.23)	−7.15

Model 1 (Nested Logit) is numerically invariant to four decimal places across all scenarios we estimated ($\hat{\alpha} = -1.078$, $\hat{\rho} = 0.574$, elasticity $= -3.55$ in every case), including three more aggressive scenarios (Belgium $L \leq 3$; Germany/Slovakia $L = 4$) that push Model 2 into poorly identified territory; these are discussed in the text (Section 6.4) rather than tabulated, since we do not view them as informative about the true model. Standard errors in parentheses for $\hat{\alpha}$; t-statistics for $\hat{\sigma}_{price}$ use robust standard errors.

Table 11: Own-Price Elasticity by Price Quartile: Model 1 vs. Model 3

Price quartile	Mean price (€)	Price ratio to Q1	$\lvert\eta_{jj}\rvert$ Model 1	$\lvert\eta_{jj}\rvert$ Model 3
Q1 (cheapest)	56.6	1.00	1.43	3.31
Q2	97.3	1.72	2.46	5.49
Q3	154.5	2.73	3.90	8.19
Q4 (most expensive)	254.3	4.49	6.43	11.81

Own-price elasticities averaged within each price quartile, across all $N = 16{,}858$ product-market observations. Model 1's elasticity-to-price ratio is constant across quartiles (within 0.1%), as implied by a single population-wide $\hat{\alpha}$ (Equation (4)); Model 3's grows sub-proportionally, reaching only 96.7%, 90.9%, and 79.6% of proportional growth in Q2–Q4 relative to Q1, mirroring Grigolon and Verboven (2014)'s Table 8.

Appendix B. Additional Figures

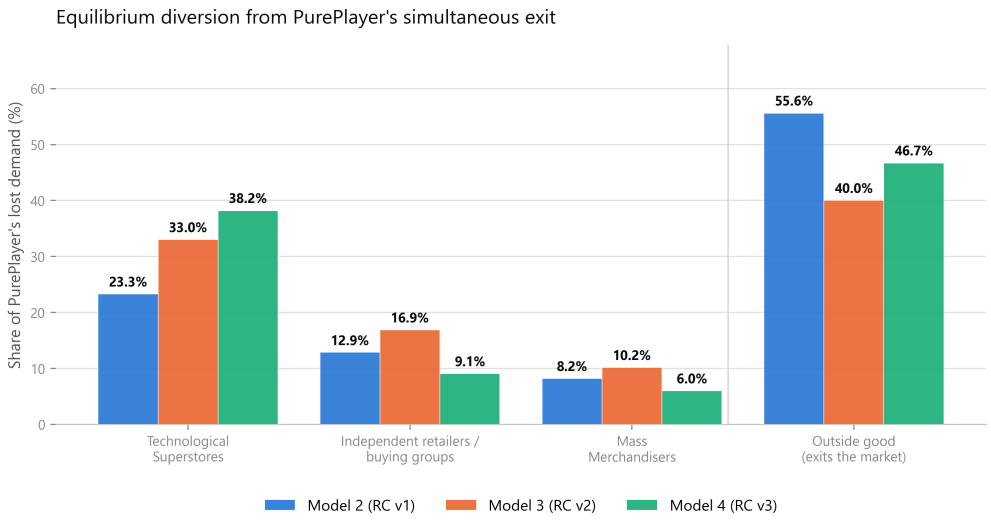


Figure 2: Equilibrium diversion ratios from the online-only channel's simultaneous exit, aggregated across all 15 markets, comparing Model 2 (RC v1), Model 3 (RC v2), and Model 4 (RC v3), $N = 16{,}858$

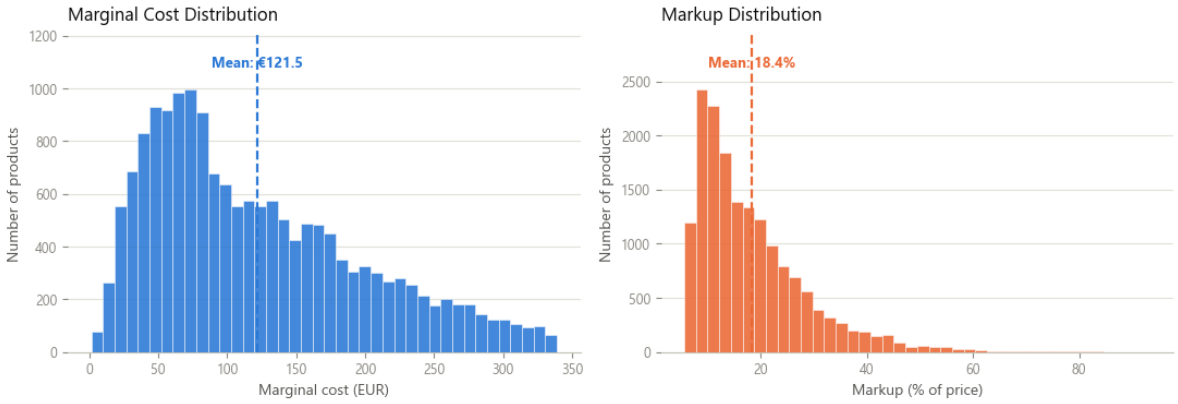


Figure 3: Distribution of marginal costs (€) and markups (% of price) at the base equilibrium, Model 2 ($N = 16{,}858$)

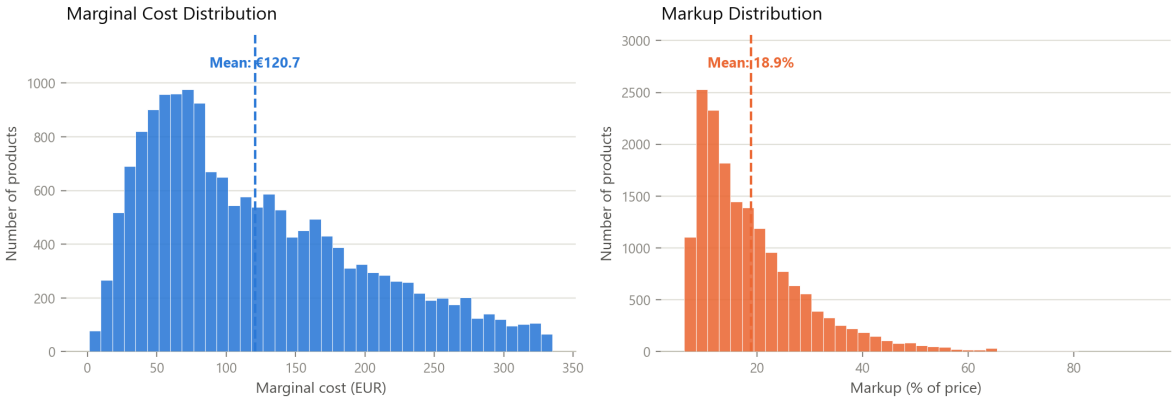


Figure 4: Distribution of marginal costs (€) and markups (% of price) at the base equilibrium, Model 3 ($N = 16{,}858$)

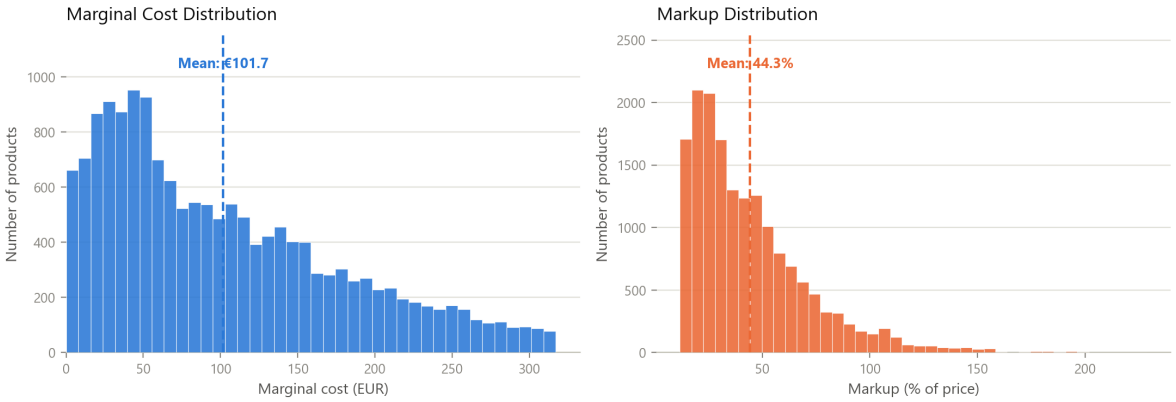


Figure 5: Distribution of marginal costs (€) and markups (% of price) at the base equilibrium, Model 4 ($N = 16{,}858$)

Appendix C. Consumer Surplus Calculation

Consumer surplus is given by the "log-sum" formula introduced by McFadden (1978) and Anderson and De Palma (1992). For the random coefficients logit model, consumer surplus is expressed as the log-sum formula of a logit, integrated over the idiosyncratic taste realizations:

$$CS = \int \frac{1}{\alpha} \ln \left(1 + \sum_{j=1}^{J} \exp(\delta_j + \mu_j(v)) \right) dP_v(v) \tag{14}$$

where the components δ_j and $\mu_j(v)$ correspond respectively to the mean utility and the individual-specific heterogeneity, and v_i represents the standard normal variable for the idiosyncratic tastes for price.

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